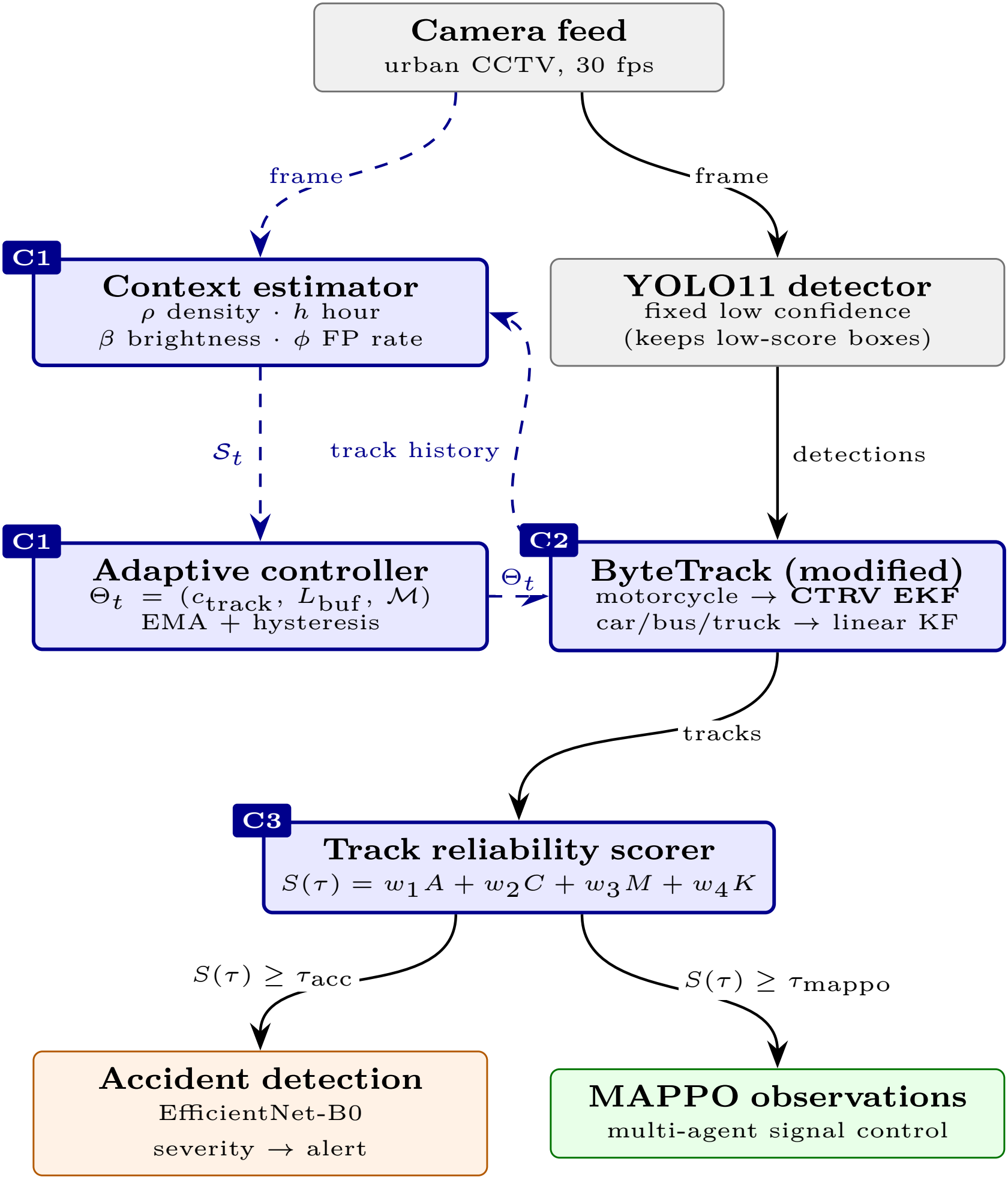
Camera feed
urban CCTV, 30 fps
frame
frame
C1
Context estimator
ρ density · h hour
β brightness · ϕ FP rate
YOLO11 detector
fixed low confidence
(keeps low-score boxes)
$\mathcal{S}_t$
track history
detections
C1
Adaptive controller
$\Theta_t = (c_{\text{track}}, L_{\text{buf}}, \mathcal{M})$
EMA + hysteresis
Θ_t
C2
ByteTrack (modified)
motorcycle → CTRV EKF
car/bus/truck → linear KF
tracks
C3
Track reliability scorer
$S(\tau) = w_1 A + w_2 C + w_3 M + w_4 K$
$S(\tau) \geq \tau_{\text{acc}}$
$S(\tau) \geq \tau_{\text{mappo}}$
Accident detection
EfficientNet-B0
severity → alert
MAPPO observations
multi-agent signal control